

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester I Examination, 2023-24

Paper No. MTB-102: Geometry

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 069661

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Find the latus rectum and the points on the conic $\ell/r = 3 - 2\cos\theta$ whose radius vector is 2.
- (b) Find the angle between the planes $x + y + z = 1$ and $2x - 3y + z = 1$.
- (c) Find the equation of the plane containing lines through the origin with direction cosines proportional to $(1, -2, 2)$ and $(2, 3, -1)$.
- (d) Show that the line $\frac{x}{1} = \frac{y-5}{3} = \frac{z+8}{-4}$ lies on the plane $x + y + z = 0$.
- (e) Name the surface $x^2 + y^2 + z^2 - x - 2y - 3z = 0$ and state four points through which it passes.
- (f) Write the equation of a sphere through the circle $x^2 + y^2 - x = 0, z = 0$.
- (g) Write the equation of a cone whose vertex is $(0, 0, 0)$.
- (h) Define right circular cylinder.
- (i) Name the surface $\frac{x^2}{4} - \frac{y^2}{9} + \frac{z^2}{16} = 1$ and tell which axis it does not intersect.
- (j) Find the equation of the normal to $3x^2 + 4y^2 + 5z^2 = 64$ at $(1, 2, 3)$.
- (k) The plane $2x - 4y - z + 3 = 0$ touches $x^2 - 2y^2 = 3z$. Find the point of contact.

Question 2.

- (a) Derive the equation of a conic in polar coordinates. [6]
- (b) Find the polar equation of the normal at any point P (vectorial angle α) of the conic $\ell/r = 1 - e\cos\theta$. [6]

Question 3.

- (a) A variable plane passes through (a, b, c) and meets coordinate axes in A, B, C . Show the locus of the point common to the planes through A, B, C parallel to coordinate planes is $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$. [6]
- (b) Show that the lines $\frac{x+4}{3} = \frac{y+6}{5} = \frac{z-3}{-2}$ and $\frac{x-3}{1} = \frac{y+5}{5} = \frac{z+1}{4}$ are coplanar. Find the plane containing them. [6]
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Question 4.

- (a) A sphere of constant radius $2k$ passes through the origin and meets the axes in A, B, C . Find the locus of the centroid of tetrahedron $OABC$. [6]
- (b) Find the sphere having the circle $x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$, $x + y + z = 3$ as its great circle. [6]
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Question 5.

- (a) Find the cone with vertex (α, β, γ) whose generators intersect $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, $z = 0$. [6]
- (b) A point P moves so that the section of its enveloping cone of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ by $z = 0$ is a circle. Show that P lies on $\frac{x^2}{a^2 - b^2} + \frac{z^2}{c^2} = 1$. [6]
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Question 6.

- (a) Find the cylinder whose generators have direction ratios $(1, 1, -1)$ and touch $x^2 + y^2 + z^2 = 4$. [6]
- (b) Find the right circular cylinder of radius 2 with axis $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{1}$. [6]
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Question 7.

- (a) Show that the locus of intersection of three mutually perpendicular tangent planes to a central conicoid is a sphere concentric with the conicoid. [6]
- (b) Show that $3x + 12y - 6z - 17 = 0$ touches $3x^2 - 6y^2 + 9z^2 + 17 = 0$ and find the point of contact. [6]
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